Curriculum Vitae

Bangya Liu
liubangya@gmail.com
neutrinoliu.github.io

Education

- **University of Wisconsin-Madison** — Madison, WI
  Doctorate, Electrical and Computer Engineering — *May. 2021 – July. 2026*
  Master of Research, Electrical and Computer Engineering, Computer Architecture track — *Sept. 2019 – May. 2021*
- **Huazhong University of Science and Technology** — Wuhan, P.R.China
  Bachelor of Engineering in Electrical and Electronics — *Sept. 2015 – Jun. 2019*

Academic Experience

- **Wisconsin Wireless and Networking Systems Laboratory** — Madison, WI
  Research Assistant, Supervisor: Prof. Suman Banerjee — *From Feb. 2021*
- **Wisconsin Computational Intelligence Laboratory** — Madison, WI
  Research Assistant, Supervisor: Prof. Jing Li — *Sept. 2019 – Dec. 2019*

Professional Experience

- **Abaka.ai** — Palo Alto, CA
  Member of Technical Staff — *Feb. 2026 – Now*
- **ByteDance E-Commerce R&D** — San Jose, CA
  PhD MLE Intern, Mentor: Dr. Xinyu Gong — *May. 2025 – Nov. 2025*
- **ByteDance Networking R&D** — San Jose, CA
  PhD SDE Intern, Mentor: Dr. Archie Abhashkumar, Dr. Weirong Jiang — *May. 2024 – Aug. 2024*
- **Telefónica Research** — Remote
  Research Intern, Mentor: Dr. Andra Lutu — *Sept. 2022 – Sept. 2023*

Publications

- **[In submission] Behavior Uncloning: Distilling Mode Redirection into Policy Weights without Inference-Time Steering**
  *Hao Wang, Jiuzhou Lei, Dayou Li, **Bangya Liu**, Minghui Zheng, Manling Li, Ruohan Zhang, Zhiwen Fan*
- **[In submission] [arXiv] ByteLOOM: Weaving Geometry-Consistent Human-Object Interactions Videos through Progressive Curriculum Learning**
  ***Bangya Liu**, Xinyu Gong, Zelin Zhao, etc., Suman Banerjee, Hao Zhang*
- **[arXiv] SwinGS: Sliding Window Gaussian Splatting for Volumetric Video Streaming with Arbitrary Length**
  ***Bangya Liu**, Suman Banerjee.*
- **[arXiv] Accelerating Transformer-Based Monocular SLAM via Geometric Utility Scoring**
  *Xinmiao Xiong†, **Bangya Liu†**, Hao Wang, etc., Yang Zhou, Zhiwen Fan.*
- **[CVPR 2026] [arXiv] SpatialStack: Layered Geometry-Semantic Fusion for 3D Spatial Reasoning**
  *Jian Zhang†, Shijie Zhou†, **Bangya Liu†**, Achuta Kadambi, Zhiwen Fan*
- **[arXiv] Egocentric World Model for Photorealistic Hand-object Interaction Synthesis**
  *Dayou Li, Lulin Liu, **Bangya Liu**, etc., Chenyu You, Zhiwen Fan*
- **[CVPR Findings 2026] [arXiv] CETCam: Camera-Controllable Video Generation via Consistent and Extensible Tokenization**
  *Zelin Zhao, Xinyu Gong, **Bangya Liu**, Ziyang Song, Jun Zhang, Suhui Wu, Yongxin Chen, Hao Zhang*
- **[SIGGRAPH 2026] [arXiv] MV-S2V: Multi-View Subject-Consistent Video Generation**
  *Ziyang Song, Xinyu Gong, **Bangya Liu**, Zelin Zhao*
- **[arXiv] VEFX-Bench: A Holistic Benchmark for Generic Video Editing and Visual Effects**
  *Xiangbo Gao, Sicong Jiang, **Bangya Liu**, etc., Qing Yin, Zhengzhong Tu*
- **[HotMobile 2026] [ACM] Privacy-Aware Sharing of Raw Spatial Sensor Data for Coop-Perception**
  ***Bangya Liu**, Chengpo Yan, Chenghao Jiang, Suman Banerjee, Akarsh Prabhakara*
- **[COLM 2025] [arXiv] Can Test-Time Scaling Improve World Foundation Model?**
  *Wenyan Cong, Hanqing Zhu, Peihao Wang, **Bangya Liu**, etc., Zhiwen Fan, Zhangyang Wang*
- **[TNET 2026] [IEEE] Sustainable Spectrum Crowdsensing**
  ***Bangya Liu†**, Yijing Zeng†, Yilong Li, Domenico Giustiniano, Suman Banerjee.*
- **[TRETS 2020] [ACM] MEG: a RISCV-based System Emulation Infra for Near-data Processing**
  *Jialiang Zhang, Yue Zha, Nicholas Beckwith, **Bangya Liu**, Jing Li.*

Selected Projects

- **OmniHaMeR: Foundation Model for Metric Hand Pose Estimation** Abaka.ai [ongoing]
  - **Keywords**: *Hand Pose, ViT, Vision Foundation Model, Embodied AI*
  - **Description**: Training a vision foundation model that ingests arbitrary inputs (RGB, depth, camera parameters, hand masks) and regresses metrically accurate hand meshes in world coordinates. Unlike RGB-only methods limited to root-aligned poses, it targets absolute MPJPE via a MapAnything-style architecture.
- **CyberArena: Adversarial Cybersecurity Arena for LLM Coding Agents** Abaka.ai [ongoing]
  - **Keywords**: *LLM Agents, Cybersecurity, LLM Arena, Benchmark*
  - **Description**: Building a live-combat evaluation environment where attacker and defender agents fight over a shared simulated sandbox under partial observability, moving beyond static CTF benchmarks. Proposed the Defend/Attack Token Ratio (DATR), the first token-level metric quantifying offense/defense economic asymmetry. Empirically tests the "cybersecurity as proof-of-work" thesis across model capability tiers.
- **ByteLOOM: Weaving Geometry-Consistent Human-Object Interactions Videos through Progressive Curriculum Learning** ByteDance [Project Page]
  - **Keywords**: *HOI, DiT, 3D Reconstruction, Wan2.1*
  - **Description**: Developed a human-object interaction video generation model based on Wan2.1 featuring realistic human dynamics and diverse object viewpoint synthesis. Addressed human-object dataset scarcity by curating large-scale open-source hand-object interaction datasets to train the DiT. To enhance controllability and geometric fidelity, introduced a novel geometry-aware conditioning mechanism combining multi-view object reference images with relative coordinate maps to support consistent generation across diverse viewpoints.
- **SwinGS: Sliding Window Gaussian Splatting for Volumetric Video Streaming with Arbitrary Length** [Project Page]
  - **Keywords**: *Gaussian Splatting, Volumetric Video*
  - **Description**: Proposed SwinGS, a novel framework for training, delivering, and rendering volumetric video in real-time streaming fashion. To address challenges including excessive model sizes, constraints on video duration, and content deviation, integrated spacetime Gaussian with Markov Chain Monte Carlo to adapt the model across various 3D scenes and frames. Employed a sliding window to capture Gaussian snapshots for each frame in a continuous training approach.
- **SpatialStack: Layered Geometry-Semantic Fusion for 3D VLM Spatial Reasoning** [Project Page]
  - **Keywords**: *VLM, 3D Spatial Reasoning, Geometry Foundation Model*
  - **Description**: Proposed a hierarchical fusion framework that aligns multi-level geometric features with language backbones for 3D spatial reasoning in VLMs. Unlike prior work that fuses only deep-layer features from vision and geometry encoders and discards rich hierarchical signals, SpatialStack progressively stacks features across multiple model depths to jointly capture fine-grained local geometry and broader contextual semantics. The resulting VLM-SpatialStack achieves state-of-the-art performance across multiple 3D spatial reasoning benchmarks.
- **MachSim: a Speedy and Scalable Flow Simulator for Traffic Distribution Prediction in Datacenter Network** ByteDance
  - **Keywords**: *Traffic Simulator, Datacenter Network, Scalability*
  - **Description**: Designed, implemented, and evaluated a scalable flow simulator for datacenter networks. The simulator models traffic distribution by taking network topology, forwarding rules, and flow information as inputs. Leveraging optimization techniques including flow as directed acyclic graph and local flow merging, together with multiprocessing, MachSim achieves high efficiency, simulating 500K flows across 100 switches in 15 seconds. Integrated into ByteDance network infrastructure to verify configuration updates before production deployment.

Honors

- **Reviewer:** Contributes 30+ reviews for NeurIPS[24][***TopReviewer*'25**], ICLR[25], ICML[25][26], AAAI[25], ACMMM[25], 3DV[25], IROS[26], IEEE TCSVT[26]
- **CVPR 2026 End-to-End 3D Learning Workshop Organizer** 2026
- **ACM 2023 S3 Workshop Program Committee Member** 2023
- **China National Scholarship** 2017, top 2%, Ministry of Education of P.R.China
- **China National Olympiad in Mathmatics in Provinces** 2014, First Price
- **China National Olympiad in Informatics in Provinces** 2014, First Price